



EU Digital Product Passport Compliance AI Agent for Pakistani Textile Exporters

Portfolio Project 2 · Junaid Iqbal · iamjunaidiqbal.com

Python 3.10	LangGraph	ChromaDB	Claude AI	Streamlit
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01 — PROBLEM STATEMENT

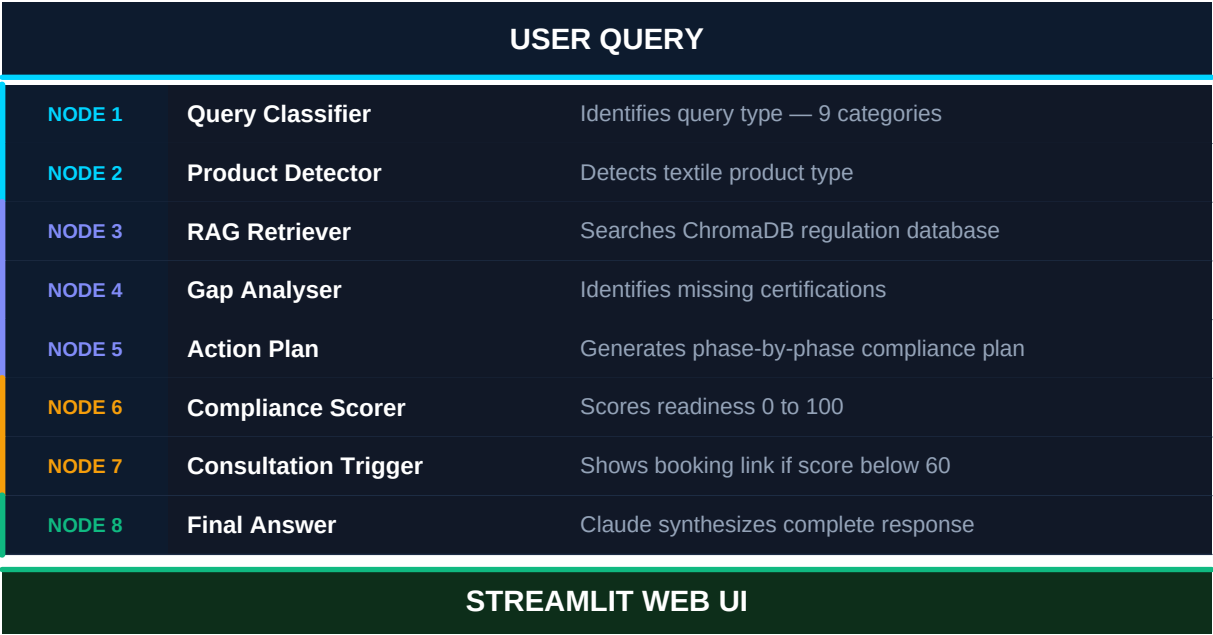
Pakistan exports **billions of dollars** of textiles to Europe every year — denim from Karachi, towels from Faisalabad, knitwear heading to Germany, France, and the Netherlands. By **2027**, every textile product entering the EU market must have a **Digital Product Passport (DPP)** — a digital record of material composition, chemical substances, carbon footprint, supplier chain, and recycling instructions. Products without a DPP will be refused at EU borders.

CHALLENGE	IMPACT
Complex EU regulations in technical language	Exporters don't know where to start
ESPR, REACH, ZDHC, OEKO-TEX requirements	Certification costs up to PKR 600,000/year
No Pakistan-specific compliance guidance	Risk of losing EU market access
2027 DPP deadline approaching fast	GSP Plus trade status at risk
Multiple labs, certifications, timelines	Overwhelming for small exporters

02 — THE SOLUTION

DPPBot is an **AI agent** that answers EU DPP compliance questions for Pakistani textile exporters in plain language. It searches a knowledge base of EU regulations, identifies certification gaps, generates step-by-step action plans with costs in PKR, scores compliance readiness from 0 to 100, and triggers consultation booking when needed.

FEATURE	DESCRIPTION
Natural language Q&A	Ask any DPP question in plain English
Pakistan-specific answers	Costs in PKR, labs in Karachi/Lahore, GSP Plus context
Compliance scoring	0-100 score with readiness level (Not Ready / Partial / Compliant)
Action plan generation	Phase by phase plan with timelines and costs
RAG-powered accuracy	Answers from real EU regulation documents, not hallucinations
Consultation trigger	Auto-shows booking link when score is below 60



DPPBot uses **Retrieval Augmented Generation (RAG)** — it searches real EU regulation documents before answering, ensuring accuracy. The knowledge base contains 6 regulation files stored in **ChromaDB** as searchable vectors using the **all-MiniLM-L6-v2** embedding model (free, local).

FILE	CONTENTS	KEY DATA
espr_regulation.txt	ESPR Article 7, DPP requirements	2027 deadline, penalties, data fields
reach_requirements.txt	SVHC substances, banned chemicals	240+ restricted substances, testing labs
zdhc_guide.txt	ZDHC certification guide	Levels 1-2-3, PKR costs, Pakistan labs
certifications_guide.txt	All EU certifications	OEKO-TEX, GOTS, GRS, Bluesign, SA8000
product_specific.txt	Product requirements	Denim, knitwear, home textiles, garments
pakistan_exporter_guide.txt	Pakistan-specific guide	GSP Plus, Karachi labs, PKR costs, mistakes

HOW RAG WORKS IN DPPBOT

STEP	ACTION	DETAIL
1	User asks a question	e.g. 'How much does ZDHC cost in Pakistan?'
2	Query converted to vector	all-MiniLM-L6-v2 converts text to 384 numbers
3	ChromaDB similarity search	Finds top 5 most relevant regulation chunks
4	Context passed to Claude	Regulation text becomes Claude's knowledge
5	Answer generated	Claude answers from real documents, not hallucination

COMPONENT	TECHNOLOGY	PURPOSE	COST
AI Model	Claude Haiku (Anthropic)	Generates answers, classifies queries	\$0.50 total
Agent Framework	LangGraph	Connects 8 nodes into pipeline	Free
Vector Database	ChromaDB	Stores and searches regulation text	Free, local
Embedding Model	all-MiniLM-L6-v2	Converts text to searchable vectors	Free, local 80MB
Web Interface	Streamlit	Chat UI and compliance dashboard	Free
Language	Python 3.10	All notebooks and app code	Free
Security	bleach library	Input sanitization against attacks	Free
Key Storage	python-dotenv	Secure API key management	Free

06 — COMPLIANCE SCORING SYSTEM

Every query triggers a compliance score based on certifications mentioned. The score drives the consultation trigger — scores below 60 automatically show the booking link.

CERTIFICATION	POINTS	WHY IT MATTERS
OEKO-TEX Standard 100	+20 pts	EU buyers demand this first — tests harmful substances
ZDHC Certification	+20 pts	Chemical management — critical for denim exporters
Bluesign	+15 pts	Process and chemical safety standard
ISO 14001	+15 pts	Environmental management system
GRS (Global Recycled Standard)	+10 pts	Required for recycled fiber products
GOTS (Organic Standard)	+10 pts	Required for organic cotton products
SA8000 (Social Accountability)	+10 pts	Growing importance for EU buyers

0 — 30	31 — 60	61 — 80	81 — 100
NOT READY	PARTIAL	MOSTLY READY	COMPLIANT

THREAT	PROTECTION	HOW
API key theft	.env file + .gitignore	Key never written in code or uploaded to GitHub
Prompt injection	bleach.clean() sanitization	HTML tags stripped from all user input
Input overflow	500 character limit	Long inputs truncated before API call
Category manipulation	Whitelist validation	Only 9 known categories accepted, rest defaulted
XSS attacks	bleach library	Dangerous characters removed on every input
Data poisoning	Input validation	All inputs validated before database queries

FILE / FOLDER	PURPOSE
app.py	Streamlit web application — chat UI + compliance dashboard
DPPBot_Notebook1_Setup.ipynb	Project setup, library install, data generation
DPPBot_Notebook2_ChromaDB.ipynb	Vector database setup and RAG pipeline
DPPBot_Notebook3_Agent.ipynb	8-node LangGraph agent construction and testing
dppbot/.env	API key storage (not in GitHub)
dppbot/data/exporters.json	30 synthetic Pakistani exporter profiles
dppbot/data/regulations/	6 EU regulation knowledge base text files
dppbot/chromadb_store/	Persistent ChromaDB vector database
dppbot/reports/	Generated compliance reports folder

3	8	6	30	13	\$0.50
Jupyter Notebooks	LangGraph Nodes	EU Regulation Files	Exporter Profiles	ChromaDB Chunks	Total API Cost

SAMPLE QUESTIONS DPPBOT ANSWERS

QUERY	RESPONSE TYPE
I export denim to Germany. What certs do I need by 2027?	Cert table + 24-month plan + PKR costs
What is ZDHC and how do I get it in Pakistan?	Step by step guide + Karachi labs + timeline
What chemicals are banned in EU textile exports?	REACH substance list + testing requirements
How much does DPP compliance cost for a small exporter?	Cost breakdown in PKR + priority order
My buyer wants OEKO-TEX. Where do I start in Karachi?	SGS/Bureau Veritas contacts + process

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